International Review of Economics and Finance 94 (2024) 103406

Contents lists available at ScienceDirect

International Review of Economics and Finance

journal homepage: www.elsevier.com/locate/iref

Does green technology innovation promote green economic growth? –Examining regional heterogeneity between resource-based and non-resource-based cities

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ARTICLE INFO

Keywords:
Green technology innovation
Green economic growth
Dual carbon
DDF
Resource-based cities

ABSTRACT

Using the radial Direction Distance Function (DDF) Golbal-Malmquist-luenberger (GML) index to calculate the growth efficiency of all-factor green economy, 262 resource-based and non-resource-based cities in China from 2003 to 2018 were taken as samples. The effects of green technology innovation on the growth efficiency of green economy and its heterogeneity are discussed through the fixed-effect model, and the internal influencing mechanism between resource dependence, green technology innovation and green economy growth in resource-based areas is further tested. The results show that: under the whole sample, green technology innovation has an "inverted U-shaped" nonlinear effect on the efficiency of green economy. Green technology innovation promotes the growth of green economy, However, with the continuous improvement of the level of green technology innovation, its marginal effect on the efficiency of green economy growth decreases. In the heterogeneity analysis of resource-based cities and non-resource-based cities, it is found that the impact of green technology innovation on green economic growth in resource-based cities is greater than that in non-resource-based cities. In the heterogeneity analysis of different resource-based cities, it is found that green technology innovation of growing and mature resource-based cities will significantly promote the growth of green economy. The mechanism test shows that the dependence of resource-based cities on natural resources will have a "crowding out effect" on human capital investment and science and technology investment.

1. Introduction

The deterioration of the global climate and environment makes it necessary to follow a strategy of green and sustainable development. With the rapid development of the global economy, the city is at the core of regional economic development. Gathering various economic information and activities and ensuring the proper distribution of factors to achieve green economic growth and green transformation and development is an essential direction of development. This is especially relevant and urgent for resource-endowed cities in which natural resource extraction and processing are the leading industries. These cities are faced with a single industrial structure, the risk of resource depletion, environmental pressure, and challenges of transformation and development.

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https://doi.org/10.1016/j.iref.2024.103406
Received 2 April 2024; Received in revised form 13 June 2024; Accepted 18 June 2024
Available online 19 June 2024
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Promoting the sustainable development of resource cities to realize green economic growth requires that we move from a single industry structure that has high energy consumption, low output, and is resource dependent (Guan, Dong, & Zhang, 2021). ESG rating, which is closely related to the pursuit of a green economy, has gradually become an essential point of reference in decision-making, even in the field of investment (Wang et al., 2024).

Green technological innovation is considered a critical factor in promoting green economic growth. Such innovation can improve resource efficiency, reduce environmental pollution, promote the upgrading of industrial structures, and transform the mode of economic development. At the same time, green technology innovation and research and development require considerable capital and human resources; while developing countries may have the required human resources, they often need more capital and technology. In addition, promoting and applying green technologies requires policy support and improved market mechanisms.

Some developing countries, such as China (in its Thirteenth Five-Year Plan), have, among other requirements, emphasized the importance of transforming and upgrading resource-based cities, strengthening the construction of an ecological civilization, and promoting green development. In December 2022, the government issued its "Implementation Programme on Further Improving the Market-Oriented Green Technology Innovation System (2023–2025)," making clear the need to vigorously develop green technologies, support green technology innovation, and improve the system of green technology innovation, highlighting its supporting role for sustainable development (Cao, 2022). What will impact green economic growth in developing countries, given the need to support green technology innovation? Is this effect heterogeneous for resource and non-resource cities? What are the mechanisms at play, and what is the relationship between green economic growth and green technology innovation under different conditions? The answers to these questions are relevant to green economic growth in developing countries.

Existing studies on the impact of green technological innovation on economic growth can be divided into four main categories. First, green technological innovation reduces enterprises' energy consumption and improves resource-allocation efficiency by promoting technologies for cleaner production, energy saving, and environmental protection, thus effectively reducing their production costs, obtaining a competitive advantage, and growing the green economy. Scholars in this line of research (Pu, Liu, & Yang, 2022) argue that green technological innovation, transformation, and upgrading positively (Zhou, Tian, & Yang, 2023) affect energy efficiency and enhance growth (Ghisetti & Quatraro, 2017).

Green technological innovation can improve enterprises' production quality (and thus promote green economic growth) by upgrading their production processes, assuming that factor inputs such as capital and labor remain unchanged (Chen, Cheng, & Dai, 2017). Urban green technology innovation can significantly improve urban green economic efficiency and promote urban green economic efficiency by reducing CO emissions and polluting manufacturing industry agglomeration(Chen & Yang, 2024).

(The presence of renewable energy, a carbon tax and GDP, green technology innovation, and environmental taxes help slow the rise of CO_2 emissions in these countries, thus promoting green economic growth (Wang et al., 2023). Green technology innovation has a positive effect on green economic growth by reducing energy intensity and promoting the upgrading of industrial structures (Li & Qin, 2022). Not coincidentally, the continuous application of green technological innovation will enhance such upgrading, thus promoting green economic growth (David and Autor, 2006).

The second line of studies treats green technology innovation as a mediating factor in the promotion of green economic growth. Market-incentivized environmental regulation stimulates green technological innovation and thus promotes green economic growth (Fan & Sun, 2020). Wang, Jiao, & Bu (2023) find that digital transformation promotes green economic growth by encouraging firms to make green technological innovations in fulfillment of their ESC responsibilities.

Che & Wang (2023) examines digital economic innovation and green economic growth in a sample of Chinese listed energy companies to examine the impact of the digital economy on green economic growth, finding that green technology innovation in alternative energy production, transport, administrative regulation, and design is an important channel for this impact. Hao et al. (2023) and Ivus & Boland (2015) examine the Fixed effect mechanism and linear relationship between digitalization and green economic growth, with their mechanism analysis showing that green technology innovation acts as a mediating variable to enhance the efficiency of green economic growth.

The third category of studies considers the heterogeneity and imbalance in the impact of green technology innovation on green economic growth. Razzaq et al. (2023) examines this impact using Chinese regional data from 2007 to 2019 and finds that green technology innovation contributes to green growth, mainly in the eastern region. Qian & Shuming (2022) measures the overall, inter-period, and sub-industry green technology innovation efficiency of 30 provinces in China from 2012 to 2019, finding that regions or provinces that are economically developed, focused on manufacturing, and with non-resource-based economies score relatively high.

The fourth category includes studies that examine the impact of green technology innovation on green economic growth as a non-linear relationship. Li & Gao (2022) finds that there is a double-threshold effect on financial development; green technology innovation will have different degrees of impact on green economic growth depending on the level of financial development. Hu & Le (2023) finds that green technology innovation has a robust spatial spillover effect, with a U-shaped non-linear relationship with green economic growth, especially in the western region.

Studies on the measurement of green economic growth from scholars within and outside of China mainly employ data envelopment analysis and the entropy value method, which requires high-quality data but ignores the importance of the indicator itself. Further, the measurement of green technological innovation relies on the number of inventions or patent applications and authorizations, a biased approach that does not allow a comprehensive evaluation of the level of green technological innovation. Most existing studies analyze the impact of green technological innovation on the green economy between provinces or countries. Few analyze the impact of green technological innovation by examining these impacts at the city level and according to whether or not they are resource reliant.

The existing literature argues that green technology innovation can positively influence green economic growth, but whether such

innovation promotes or crowds out green economic growth and the path of any impact remains unclear. This study responds by contributing in the following ways. First, we use the Golbal-Malmquist-Luenberger (GML) index – based on the radial directional distance function (DDF) –to calculate the total factor green economic growth rates of 262 resource and non-resource cities in China from 2003 to 2018.

Second, based on the OECD's 2012 OECD Green Growth Monitoring Indicator System" and the available data indicators, we select six positive indicators and use the entropy value method to construct a system to evaluate green technological innovation using comprehensive indicators. Third, this paper analyses the heterogeneity of resource and non-resource-based cities and empirically examines how the crowding-out effect in resource-based cities affects green technological innovation and green economic growth. Fourth, following the empirical analysis, we find that green technological innovation and growth efficiency of the green economy are in a non-linear relationship, which is a revelation for improving the system of green technological innovation and promoting the green transformation of resource-based cities. Fifth, with the rapid development of the global economy, environmental issues have become increasingly prominent, and sustainable development has become the focus of common concern of all countries. As a new and environmentally friendly way of economic development, green technology innovation is considered to play an important role in promoting green economic growth. From an international perspective, cities in different countries differ greatly in terms of resource endowment, industrial structure and technological level, and how these differences affect the relationship between green technology innovation and green economic growth is a question worthy of in-depth discussion. Through the comparison and analysis of different countries, it can provide beneficial enlightenment and reference for the development of green economy in the global scope.

2. Literature review and research hypotheses

2.1. Green technological innovation and green economic growth

Scholars in China and elsewhere have considered how green economic growth is defined and assessed. The concept of the green economy was first advanced by the United Nations Conference on Sustainable Development in 2012; the conference also promoted the synergistic development of economic growth through the development of ecological civilization, ecosystem construction, and building ecological awareness as the core of the research. Green economic growth goes beyond traditional notions of economic development to include a focus on equity in social opportunity, green production and consumption, and ecological environmental protection (Zhou, 2018); these factors are considered to more accurately reflect a country or region's level of economic development. However, there is currently no authoritative definition of green economic growth.

In its 2011 report on the "Green Growth Strategy," the OECD considers subjective human factors, proposing that, from the perspective of individual satisfaction, green growth is an important channel to promote economic growth and development through the rational use of natural resources and access to a variety of factors and environmental services. Brown et al. (2014) argue that the green economy can change how economies operate and fundamentally change people's attitudes, thereby promoting stable long-term economic development.

Loiseau argues that the top priority of green economic growth is to pay attention to the synergistic development of reasonable consumption of resources and the development of the economy (Loiseau et al., 2016). Wang, Wang, & Zhang (2024) believes that research on the green economy needs to be extended to the macroeconomic field and take an international perspective based on traditional trade and foreign direct investment. Further, foreign aid should be incorporated into the framework for research on the carbon emissions impact factor.

In evaluating and measuring green economic growth, research has focused on the multi-factor non-parametric evaluation method; for example, Chen et al. (2018) evaluates the green total factor growth of the three major city clusters in the Yangtze River Economic Belt using the super-efficient SBM model to measure their level of green economic growth. Li (2021) empirically measures the evolution of spatiotemporal patterns of green economic growth of manufacturing industries in China's interprovincial space using the input-output Super-SBM model. Meng and Zhang (2020) uses green total factor productivity (GTFP) and energy and environmental efficiency (EEE) as measures of the efficiency of green economic growth, constructing and measuring GTFP and EEE indices using the non-radial DDF method.

Some scholars use green GDP to measure green economic growth; Yang & Shan (2019) uses the PCF method to measure regional ecological science and technology innovation capacity and constructs the regional green GDP accounting formula to analyze the level of green economic growth in each region. Zhao et al. (2022) uses the super-efficient data envelopment analysis (DEA) model to measure the level of green economic growth in Chinese cities. The entropy power method is used to construct evaluation indicators in (Xie et al., 2019), and based on an international comparison, the author emphasizes that the OECD's green growth framework focuses on the coordinated development of the economy and the environment and the integration of the OECD's green growth strategy framework. The author summarizes the internal logic of green growth from several perspectives, reflecting the environmental efficiency of production and consumption, the natural resource base, governmental control, and the living environment, and uses the extreme entropy power method to construct indicators to evaluate green economic growth.

The weight method is used to construct authoritative indexes to evaluate green economic growth. Cheng et al. (2019) analyzes the spatial and temporal evolution of green development in 30 provinces and municipalities in China and constructs a system of indicators of green economic growth to reasonably evaluate the level of green economic growth of provinces and municipalities; this system includes three system layers, nine factor layers and 30 indicators of green growth, green welfare, and green wealth.

At present, green technological innovation has attracted the attention of scholars within China and elsewhere. The level of green technological innovation is measured using two main approaches. First, several studies use the number of green patent applications

and authorizations. Li, Liu, & Liang (2024) uses the number of green patent applications per 10,000 people in the region to measure green technological innovation. Li, Cao, & Zhou (2023) argues that the number of green patent applications only partially reflects the importance of green technology; therefore, the logarithm of the number of authorized green invention patents is used to measure the level of green technological innovation. Zhang et al. (2018) selects the number of green patent authorizations and the proportion of green technology-related revenues to total revenues in the national high-tech industrial development zones as indicators of green innovation and green technology innovation.

Second, by measuring the efficiency level, Jiang et al. (2021) use dynamic DEA with non-expected output to measure the green technology innovation efficiency of renewable energy enterprises. Qian, Wang, & Xiao (2018) evaluate and measure the green innovation efficiency of industrial enterprises at the inter-provincial level through a two-stage DEA model, finding that the level of green innovation efficiency of industrial enterprises was low, but the results are excellent with respect to green transformation. Hu & Le (2023) uses green innovation efficiency as a proxy variable for green technology innovation based on the stochastic frontier model SFA. Fang (2023) measure green technology innovation efficiency and explore the association between green investment and green technology innovation efficiency.

Green technology innovation is an essential tool for promoting growth efficiency of the green economy. However, in the early stages of development, this innovation has limited impact on growth efficiency of the green economy because it involves high investment, high risk, and a long payback period. With government policies such as "green development" and "peak carbon and carbon neutrality," enterprises have gradually realized the importance of green technology innovation in enhancing their market position, reducing environmental management costs, and increasing investment in green technology research and development. This has led to economies of scale being established in green industrial technologies and increasing marginal returns, which has, in turn, contributed to green economic growth.

However, according to Muldaur's theory of cyclic causal accumulation, green economic growth is a dynamic socio-economic process once it reaches a certain scale. The green economic growth of all relevant factors is connected, and these are in a relationship of mutual influence and mutual causation. Green technological innovation and the interactive effect of green technological innovation and economic growth will give rise to the phenomenon of "the strongest will always be the strongest," which may lead to market monopoly, misallocation of resources, and waste. In addition, the government may also need to fine traditional polluters to promote green development (in addition to promoting innovation). These two aspects of government action work together to encourage green technological innovation.

They also prompt enterprises to increase their green innovation efforts to reduce production costs. With increased investment in green technological innovation, the marginal returns on resources are increasing, and firms can reduce their production costs by slightly improving production technology and reducing pollutant emissions, thereby promoting green economic growth. However, as resources are invested and production processes improved, the marginal returns to green technological innovation gradually diminish. If there is no transformative innovation, standard outputs may be crowded out by green technology innovation and even lead to a negative correlation between green technology innovation and green economic growth. However, the mismatch between the introduction of green technology and the resource endowment of resource cities may lead to decreased efficiency in resource utilization and a decline in the intensification of green economic growth (Tang, Fu, & Wang, 2014). Based on the above analysis, we put forward the following hypotheses:

Hypothesis 1. The relationship between green technological innovation and green economic growth is an "inverted U" curve; that is, when the level of green technological innovation reaches a certain level, marginal effect on the growth efficiency of green economy will gradually diminish.

2.2. Crowding-out of green technology innovation in resource and non-resource cities

In analyzing the relationship between green technological innovation and economic growth, generalizing from a national sample is unhelpful. The inherent characteristics of resource-based cities impact the level of green innovation of cities that are not resource dependent, leading to inconsistent efficiency in green economic growth. Therefore, this study offers a theoretical analysis of the rationale and mechanism for this effect.

Resource-based cities are those cities that are highly dependent on resource extraction and processing, resulting in resource price volatility having a significant impact on their economic development. When resource price volatility is high, the profits of resource- and energy-consuming sectors drop, reducing investment in green technology innovation (Li & Xu, 2018). This is because green technology innovation often requires long-term investment and has unstable returns. Therefore, in the case of high resource-price volatility, enterprises are more inclined to use their capital for their traditional business to ensure more stable short-term returns.

Second, resource-based cities are prone to neglect environmental protection and sustainable development in the development process due to their over-dependence on particular resources (Xu et al., 2023). As a result, these cities may lack incentives to promote green technological innovation despite the challenge of achieving sustainable development. Again, resource-based cities may fall prey to the "resource curse." The dominant indicator that is used in current government assessment is still the GDP growth rate. However, the input cycle of green technology innovation is long and slow, leading to the possibility that the government may neglect green technology innovation and thereby slow the growth of the green economy. This effect is pronounced in resource-based cities, which are primarily concentrated in the central and western regions, with relatively low levels of economic development and openness to the outside world, a lack of new industries that drive high-quality economic development, and GDP growth rates that depend on the output of the resource sector. As a result, the government invests substantial resources in the sector, including resources that could be used to

invest elsewhere in the economy (Si & Mao, 2019), which may crowd out green innovation in other sectors. Based on the above analysis, we offer the following hypothesis:

Hypothesis 2. Resource-based cities crowd out input of green innovation elements from other sectors by investing too many resources in the resource-based sector.

3. Study design

3.1. Sample and data collection

This study examines resource- and non-resource-based prefecture-level cities in China using data from the China Urban Statistical Yearbook, China Local Statistics Bureau, and China National Bureau of Statistics for 2003 to 2018, a period chosen for reasons of data availability. The data for some missing years are processed by mean and linear interpolation. We obtain final panel data for 262 prefecture-level cities in China.

3.2. Variable definition and measurement

Explained variables: Using the radial direction distance function Golbal-Malmquist-luenberger (DDF-GML) index to calculate the all-factor green economy growth efficiency value GTFP, that is, the change rate of the all-factor green economy growth rate. According to the GML index calculation method, the all-factor green growth rate of the base period is selected as 1. A value greater than 1 indicates an increase in the growth rate of all-factor green economy, while a value less than 1 indicates a decrease in the growth rate of all-factor green economy. we follow the usual practice of creating input indicators, including indicators of human input measured by the employed urban population, which we calculate as the total number of employees in primary, secondary and tertiary industries. Capital input is measured by the total investment in fixed assets, and energy consumption is measured by annual electricity consumption. Desired output is measured by gross regional product (GRP), and undesired output is measured by emissions of urban industrial wastewater, industrial sulfur dioxide, and industrial smoke and dust (see Table 1).

Core explanatory variables: This study comprehensively evaluates regional green innovation using the entropy weighting method; a total of six positive indicators are taken to calculate the level of green technological innovation in a city. These are the full-time equivalence of R&D personnel, R&D internal expenditure, and the annual number of green invention applications (*Ingrva*), green utility model applications (*Ugrma*), green inventions authorized (*Ingrvg*), and green utility models authorized (Ugrmg) (see Table 2).

Control variables: The following variables that may affect green technological innovation and green economy growth efficiency are selected to accurately assess the impact of the former on the latter and to construct the model employed: GDP per capita (*rgdp*); population size (*POP*), measured using the number of residents; industrial structure (*DECY*), measured using the proportion of secondary-industry output to GDP; the degree of marketization (*lnSCJG*), measured by the share of regional private sector employment in total regional employment; the degree of government intervention (*GOV*), taken as the ratio of government budgetary expenditures to GDP; the degree of openness to the outside world (*FDI*), taken as the sum of the current year's imports and exports as a share of the current year's GDP; and the level of financial development (*FDS*), taken as the sum of the end-of-the-year financial institutions' deposits and loans as a share of GDP (Wang, Mao, & Wu, 2023).

3.3. Modeling

According to the research hypothesis, it is necessary to verify the relationship between green innovation and growth efficiency of the green economy. Therefore, this paper tests the nonlinear relationship between green innovation and green economy growth efficiency and builds the following model.

$$GTFP_{it} = \alpha_0 + \alpha_1 GI_{it} + \alpha_2 GI_{it}^2 + \alpha_3 X_{it} + \mu_i + \varepsilon_{it} \tag{1}$$

Table 1
Indicator selection and data processing of GTFP variables.

Variable type	Variable name	Selection of indicators and description of data
Inputs	Manpower inputs	Human input is measured by the urban employed population, taken as the total number of people employed in the primary, secondary, and tertiary sectors.
	Capital investment	Total investment in fixed assets, in billions of dollars.
	Energy inputs	Energy consumption is measured by annual electricity consumption in million kWh.
Expected outputs	GDP	Gross regional product (GRP)
Non-expected outputs	Municipal industrial wastewater discharges	Adopting industrial wastewater in the "three wastes" discharged by each prefecture-level city in 10,000 tonnes.
	Industrial sulfur dioxide emissions	Adopting industrial sulfur dioxide emissions from the "three wastes" of each prefecture-level city in tonnes.
	Industrial smoke and dust emissions	Adoption of industrial fume and dust in the "three wastes" emissions of each prefecture-level city in tonnes.

Source: Compiled by the author

Table 2
Index for evaluation of green technology innovation.

Primary indicators	Secondary indicators	Tertiary indicators
Green Technology Innovation Index	Green innovation inputs	Full-time equivalent of R&D personnel
		R&D internal expenditure
	Green innovation outputs	Number of green patent applications
		Number of green patents granted
		Number of green utility model applications
		Number of green utility model authorizations

Source: Compiled by the author

where *GTFP city* is the growth in GTFP measured using the GML index, α is the parameter to be estimated, green technological innovation (*GI*) is the green innovation index, GI^2 is the squared term of the green innovation index, X_{it} is a vector of control variables, μ_i is used to control for individual heterogeneity that does not vary over time, and ε_{it} is the residual term.

4. Empirical analysis

The descriptive statistics for the main variables are shown in Table 3 below. The total number of observations is 4,512, and the sample size, mean, variance, maximum, and minimum values of each variable are set out separately.

The Pearson's test is employed to test for possible multicollinearity. The correlation coefficient matrix indicates that the absolute values of the correlation coefficients are between 0.002 and 0.433, which is less than the critical value of 0.7, indicating that there is no significant multicollinearity in the model. We ensure that the model is reliable and avoids pseudo-regression by conducting a short panel unit root test using the IPS test; all variable values are significant after first-order differencing. Given the existence of first-order unitary variables mentioned above, we explore whether there is a long-run cointegration relationship among the variables, using the Kao test to conduct the cointegration test; the ADF test results reject the original hypothesis that there is no cointegration relationship.

We construct a model for static panel data; whether to include random or fixed effects depends on the results of the Hausman test. The p-value of the Hausman test result is 0.0314, and the original hypothesis can be accepted; that is, the estimation results are better for the fixed-effect than the random model. Therefore, we adopt fixed-effect for the full-sample regression (Wang, Wang, & Zhang, 2024). The regression results are shown in Table 4.

The regression results in Table 4 show that Model (1) represents the relationship between the main explanatory variables, green technological innovation (*GI*), the square of green technological innovation (*GI2*), and growth efficiency of the green economy (GTFP). The coefficients of *GI* are significantly positive at the 1% level of significance, and the coefficients of *GI2* are significantly negative at the 1% level of significance, indicating that green technological innovation and growth efficiency of the green economy have a non-linear inverted U-shaped relationship. offering an initial verification of the hypothesized relationship and that, to a certain extent, growth efficiency of green economy shows a marginal decreasing trend, that is, green economy is increasing at a decreasing rate. The explained variable in this paper is growth efficiency of all-factor green economy GTFP, that is, the change rate of the growth rate of all-factor green economy. The negative correlation between the square term of green technology innovation, the core explanatory variable, and growth efficiency of green economy shows that with the improvement of green technology innovation, the growth efficiency of green economy slows down, but still continues to grow.

Models (2)–(8) are derived from the gradual increase of control variables based on Model (1). The results show that no matter how many control variables are added, there is a clear inverted U-shaped relationship between green technological innovation and growth efficiency of green economy, indicating that the model has a certain degree of stability. Adding control variables, we find that GDP per capita and green economic growth are negatively correlated in the full sample, reflecting that the increase in GDP per capita does not promote growth efficiency of green economy. This may be explained by the pressure on resources and the environment; that is, in developing countries like China, the increase in GDP per capita comes at the cost of massive resource consumption and environmental damage, thus limiting growth efficiency of green economy.

It is also possible that there are technological constraints: although an increase in GDP per capita may mean an increase in the level

Table 3
Descriptive statistics and analysis.

Variable	N	Mean	SD	Min	Max
GTFP	4512	1.016	0.048	0.574	1.779
GI	4512	0.014	0.040	0.000	0.745
GI^2	4512	0.002	0.016	0.000	0.555
rgdp	4512	3.607	3.084	0.010	46.775
POP	4512	43.720	30.821	1.637	340.400
DECY	4512	0.484	0.110	0.027	0.910
lnSCJG	4512	12.506	0.997	9.516	16.069
GOV	4512	0.162	0.092	0.031	1.485
FDI	4512	0.003	0.003	0.000	0.045
FDS	4512	1.648	0.542	0.141	11.827

Table 4
Full sample regression results (random-effects model).

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	GTFP	GTFP	GTFP	GTFP	GTFP	GTFP	GTFP	GTFP
GI	0.148***	0.190***	0.217***	0.262***	0.247***	0.257***	0.249***	0.251***
	(0.023)	(0.032)	(0.041)	(0.050)	(0.052)	(0.052)	(0.051)	(0.053)
GI^2	−0.271***	−0.327***	−0.363***	−0.438***	−0.411**	−0.433**	−0.419**	−0.425**
	(0.096)	(0.108)	(0.120)	(0.167)	(0.166)	(0.169)	(0.168)	(0.170)
rgdp		−0.001**	−0.001**	−0.001***	−0.001***	−0.001***	−0.001***	−0.001***
		(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
POP			−0.336*	−0.310	−0.492**	−0.456**	−0.420**	−0.430**
			(0.186)	(0.189)	(0.213)	(0.208)	(0.211)	(0.212)
DECY				0.015**	0.016**	0.021***	0.021***	0.020***
				(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
lnSCJG					0.001	0.001	0.001	0.001
					(0.001)	(0.001)	(0.001)	(0.001)
GOV						0.018**	0.019***	0.019***
						(0.007)	(0.007)	(0.007)
FDI							0.285	0.295
							(0.209)	(0.211)
FDS								6.940
								(11.242)
I.year	/	/	/	/	/	/	/	/
I.city	/	/	/	/	/	/	/	/
_cons	1.014***	1.016***	1.018***	1.011***	0.996***	0.989***	0.991***	0.990***
	(0.001)	(0.001)	(0.001)	(0.003)	(0.011)	(0.012)	(0.012)	(0.012)
N	4512	4512	4512	4209	4209	4209	4209	4209

Standard errors in parentheses.
$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

of economic development, it does not mean that the level of technology has increased accordingly. If there is a lack of advanced environmental protection technology, then an increase in GDP per capita cannot lead to growth efficiency of green economy. Therefore, the promotion of growth efficiency of green economy needs to take into account the pressure on resources and the environment, the improvement of technological innovation, and many other factors, which are precisely the problems that developing countries need to solve.

There is also a significant negative correlation between population size and growth efficiency of green economy, indicating that the increase in China's population has not motivated growth efficiency of green economy. This may be because the increase in population size has not increased human capital in sufficient time to effectively promote scientific and technological innovation, hindering growth efficiency of green economy. It is also possible that the increase in population has led to a corresponding increase in the demand for natural resources, resulting in the overexploitation of resources and energy consumption, placing a significant burden on the environment and negatively affecting growth efficiency of green economy. Developing countries need to increase investment in human capital, accelerate the transformation and upgrading of industrial structure, and further scientific and technological innovation to boost sustained growth efficiency of green economy. The industrial structure can significantly drive growth efficiency of green economy, indicating that China's industrial structure is being upgraded effectively to promote growth efficiency of green economy. Government intervention also has a significant role in promoting growth efficiency of green economy, indicating that the Chinese government's intervention effectively promotes growth efficiency of green economy, reflecting the state's attention to environmental protection and even its contribution to addressing global environmental issues. A city's degree of marketization, openness to the outside world, and level of financial development do not significantly affect growth efficiency of green economy.

5. Heterogeneity analysis

5.1. Heterogeneity analysis of resource-based and non-resource-based cities

According to the theory proposed above, analyzing a national sample on its own is not possible. A detailed analysis of the relationship between green innovation and growth efficiency of green economy requires a heterogeneity analysis that considers differences between resource-based and non-resource-based cities. The classification criteria are based on the National Sustainable Development Plan for Resource-based Cities (2013–2020) and the list of resource-depleted cities published by the State Council of China to classify cities as resource-based or not. This study offers the first definition of 262 resource-based cities in China and classifies four types: growth, maturity, decline, and regeneration cities, according to differences in their resource security and sustainable development capacity and according to different resource types (coal cities, petroleum cities, non-ferrous metal cities, and black metal cities). The classification of resource cities is complicated, with many types and noticeable regional differences, so we adopt a double-fixed model regression for resource-based cities. In contrast, the differences among non-resource cities are smaller, and random effect analysis is adopted for these. The related models include:Columns (1) and (2) are fixed-effect model of resource-based region; Columns(3) and (4)

are fixed effects modeling of non-resource regions; Columns (5) and (6) are random effects model of non-resource regions. The regression results are shown in Table 5 below.

From the regression results, there is an inverted U-shaped relationship between green technology innovation and growth efficiency of green economy in resource and non-resource cities. However, the coefficient for the variable capturing the level of green technological innovation in resource-based cities is much larger than that for non-resource-based cities. This reflects the fact that in the early stages of the application of green technology, the green economic benefits are higher in resource-based cities than in non-resource-based cities. This may be because the economic growth of the former relies more heavily on resource-processing industries. The application of green technological innovation can improve the efficiency of resource utilization, increasing economic benefits. However, the application of green technological innovation can reduce pollutant emissions, improving green economic growth. Therefore, in the early stages, the marginal benefits of green technology innovation are greater in resource-based than non-resource-based cities. However, after a particular stage of development, both resource-based cities and non-resource-based cities show marginal decreasing growth efficiency of green economic as a bottleneck develops with respect to innovative technology, the role of green technology innovation in promoting the growth of green economy tends to be stable, and there is a lack of new economic growth poles. After the development of green technology innovation to a certain extent, the growth efficiency of green economy appears marginal decline.

5.2. Analysis of different types of heterogeneity in resource-based cities

Because of the different types of resource cities, there are differences in their respective characteristics, which may lead to differences in the results according to type. We analyze the crowding-out effect for cities with different resource types in further detail by dividing cities according to the standard of the Plan as growth, maturity, decline, and regeneration cities; the regression results are shown in Table 6 below.

The results show that there is a significant "inverted U-shaped" relationship between green technology innovation and growth efficiency of green economy in growing resource cities, indicating that the marginal decreasing trend of green economy growth efficiency in growing resource cities is more obvious, This may be related to the characteristics of growing resource-based cities. Growing resource-based cities are emerging resource-based cities with short time for resource exploitation and urban construction, and various industries mainly focus on resource product exploitation. If too much resources are invested in green innovation, the expected output of growing resource-based cities will be reduced, thus showing an "inverted U-shaped" relationship.

In mature resource cities, the coefficient of the primary term is significantly positive, indicating that green innovation positively impacts growth efficiency of green economy. In contrast, the coefficient of the secondary term is not significant, indicating that these cities are in the left half of the "inverted U" curve. At this time, the growth efficiency of green economy is still in the marginal increasing stage because these mature resource-based cities are in a stable stage of resource development, with a solid ability to guarantee resources and a high level of economic and social development. Their economic growth does not depend solely on the development of resources. Their stable resource development, resource security, high level of economic and social development, and economic growth are not solely dependent on resource development and processing of low value-added and high-pollution industries.

Both the primary and square coefficients of declining resource-based cities and regenerative resource-based cities are not

Table 5
Heterogeneity analysis of resource and non-resource cities.

	(1)	(2)	(3)	(4)	(5)	(6)
GI	1.529**	2.239***	0.099**	0.228**	0.139***	0.215***
	(0.663)	(0.813)	(0.047)	(0.097)	(0.024)	(0.060)
GI^2	−14.954*	−23.419**	−0.228**	−0.399*	−0.258***	−0.376**
	(8.833)	(11.126)	(0.112)	(0.211)	(0.096)	(0.173)
rgdp		−0.001		−0.002***		−0.001***
		(0.001)		(0.001)		(0.000)
POP		0.474		0.583		−0.469**
		(7.217)		(3.333)		(0.237)
DECY		0.091*		0.020		0.014*
		(0.049)		(0.026)		(0.008)
lnSCJG		0.003		−0.001		0.002
		(0.002)		(0.003)		(0.001)
GOV		0.077**		0.071***		0.028***
		(0.030)		(0.022)		(0.009)
FDI		1.163*		−0.418		0.310
		(0.633)		(0.414)		(0.240)
FDS		−23.673		−19.882		2.244
		(46.976)		(42.770)		(17.585)
_cons	1.013***	0.928***	1.016***	1.017***	1.016***	0.987***
	(0.002)	(0.040)	(0.001)	(0.030)	(0.001)	(0.016)
N	1680	1569	2832	2640	2832	2640

Standard errors in parentheses.
*$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

Table 6
Regression results of crowding-out effects for different resource types of cities.

	(1)	(2)	(3)	(4)
	growth-oriented	mature	recessionary	regenerative
GI	7.726***	1.602*	7.062	1.885
	(2.354)	(0.882)	(6.160)	(1.713)
GI^2	−20.115***	−15.692	−7.980	−18.771
	(173.764)	(11.240)	(259.670)	(32.406)
_cons	1.165***	1.005***	1.177***	1.027***
	(0.057)	(0.009)	(0.097)	(0.104)
controls	/	/	/	/
i.year	/	/	/	/
i.city	/	/	/	/
N	208	975	329	210

Standard errors in parentheses.
*$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

significant. It is speculated here that the backward industrial structure and low level of economic development of declining resource-based cities may lead to the insignificant effect of green innovation on the growth rate of green economy. The reason why the regression result of renewable resource-based cities is not significant may be because the renewable resource-based cities have just got rid of resource dependence, the development level of new industries is at a low level, and the new economic growth path of each city is not clear, leading to the insignificant effect of green innovation on the growth efficiency of green economy.

6. Robustness and endogeneity test

6.1. Robustness test

In order to further verify the impact of green technology innovation on the growth efficiency of green economy, the robustness of the regression results of the model was tested by removing extreme data, adding model control variables and changing regression methods. The results are shown in Table 7 below.

1. Eliminate extreme variables. In Table 7, model (1) and model (2) are used to indent the main explanatory variables and the explained variables by 1% up or down. Report results for control variables are not listed due to space constraints. The regression results show that the efficiency of green technology innovation and green economy growth still meet the nonlinear "inverted U-shape" after tail-shrinking treatment of the core variables, which is consistent with the previous results, and the regression results are robust.
2. Add model control variables. Sun, Liu, & Zhou (2014) believes that the development of the tertiary industry can promote the green economy, and this paper adds the variable of the proportion of the output value of the tertiary industry in GDP (INDU) for regression. The regression results of model (3) and model (4) show that after adding other control variables, green innovation still presents a significant "inverted U-shaped" feature on the growth efficiency of green economy.

Table 7
Robustness test.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	GTFP_w		GTFP		GTFP		GTFP	
GI_w	0.176***	0.402***						
	(0.026)	(0.076)						
GI^2_w		−1.328***						
		(0.429)						
GI			0.083**	0.251***	0.179***	0.177**	0.085**	0.251***
			(0.036)	(0.056)	(0.027)	(0.088)	(0.038)	(0.071)
INDU			0.006	0.001				
			(0.018)	(0.018)				
GI^2				−0.425**	−0.321***	−0.536***		−0.425**
				(0.173)	(0.112)	(0.197)		(0.187)
L.TFPCH					−0.163***	−0.216***		
					(0.061)	(0.054)		
_cons	0.983***	0.990***	0.980***	0.990***	1.180***	1.144***	0.982***	0.990***
	(0.009)	(0.010)	(0.013)	(0.013)	(0.062)	(0.066)	(0.018)	(0.018)
N	4209	4209	4206	4206	4064	3930	4209	4209

Standard errors in parentheses.
*$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

3. Use different models for estimation. In Table 7, model (5) and model (6) adopted the system GMM method for regression, and model (7) and model (8) adopted pool OLS regression. Two different estimation methods have obtained an "inverted U-shaped" relationship between green innovation and green economic growth efficiency, indicating that this conclusion is robust.

6.2. Endogeneity test

Due to missing variables or errors in the measurement of the development level of green economy, there may be bias in the final result and thus endogenous problems. In order to solve the endogenous problem, this paper uses the first-order lag of green technology innovation as an instrumental variable for analysis, and the regression results are shown in Table 8 below.

The regression results in Table 8 show that when the lag of green technology innovation is taken as an instrumental variable, the relationship between green technology innovation and green economic growth efficiency still satisfies the "inverted U-shape" after the endogenous problem is corrected. Moreover, the F value of the first stage is 1316.97, and the F statistic is greater than the empirical value 10, which further indicates that the variable of green technology innovation lagging behind the first stage is not a weak instrumental variable.

7. Mechanism analysis

In summary, the empirical analyses show that the level of green innovation in resource- and non-resource-based cities has a marginal decreasing effect on growth efficiency of green economy after a certain level of development. However, for resource-based cities, are there other factors crowding out green innovation, and what is the mechanism for this? We explore whether there is a crowding out of other factors by green innovation in resource cities by conducting a mechanism test and selecting human capital investment (*EDU*), calculated as the proportion of year-end urban education expenditure to GDP, since the resource sector's relatively low demand for human capital may affect the region's ability to carry out technological innovation. Second, we select science and technology (*STI*) as the proportion of year-end urban education expenditure to GDP. The second choice is science and technology investment (STI), calculated as the share of science and technology expenditure in GDP at the end of the year. Regarding science and technology investment in R&D, the "resource curse" afflicting resource cities with natural resource endowments will prompt a biased choice by enterprises, prioritizing the development of resource-based industries and increasing the investment in R&D of resource-based industrial technology. Environmental regulation policy may hamper the level to which enterprises develop green technology innovation, and the R&D investment in the innovation sector is squeezed out. Therefore, we establish the following panel model:

$$GI_{it} = \alpha_0 + \alpha_1 minging_{it} + \mu_{it} + \lambda_{it} + \varepsilon_{it} \tag{2}$$

$$GI_{it} = \beta_0 + \beta_1 Z_{it} + \mu_{it} + \lambda_{it} + \varepsilon_{it} \tag{3}$$

$$Z_{it} = \gamma_0 + \gamma_1 mining_{it} + \mu_{it} + \lambda_{it} + \varepsilon_{it} \tag{4}$$

where Z_{it} denotes human capital investment, science and technology R&D investment, respectively, mining denotes the degree of resource dependence, which is expressed by the proportion of the population engaged in the extractive industry, μ_{it} and λ_t denote the fixed individual differences and time trend, respectively, and ε_{it} is the disturbance term. The regression results are shown in Table 9, Table 10 and Table 11, which respectively represent the "crowding out effect" of resource dependence on green innovation and its mechanism test.

The regression results in Table 9 analyze formula 3. Model (1) shows that resource dependence has a negative effect on green innovation level under the condition of full sample; That is, the higher the degree of resource dependence, the lower the level of green technology innovation, which is consistent with the previous analysis. Model (2) represents the impact of resource dependence on green innovation level of resource-based cities. minging coefficient is significantly negative at 1% significance level, indicating that resource dependence of resource-based cities will have a negative impact on green innovation of resource-based cities. This may be because resource-based cities put too much resources into the resource sector, which affects the efficiency of green innovation in other sectors and produces a crowding out effect. Model (3) The minging coefficient of non-resource-based cities is negative but not

Table 8
Endogeneity test.

	(1)	(2)
	IV estimate	
	GTFP	GTFP
GI	0.102*** (0.035)	
GI^2		−0.264* (0.136)
L.GI		0.223*** (0.083)
_cons	0.975*** (0.019)	0.979*** (0.019)
N	3930	3930

Standard errors in parentheses.
*$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

Table 9
The impact of resource dependence on green technology innovation.

	(1)	(2)	(3)
	GI	GI	GI
mining	−0.145*	−0.038***	−0.328
	(0.075)	(0.014)	(0.274)
_cons	0.015***	0.005***	0.020***
	(0.002)	(0.001)	(0.003)
N	4239	1663	2576

Standard errors in parentheses.
*$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

Table 10
Mechanism test - crowding-out effect on human capital investment.

	(1)	(2)	(3)	(4)
	Resource-based Regions		Non- resource-based Regions	
	EDU	GI	EDU	GI
mining	−0.113***		−0.320***	
	(0.032)		(0.084)	
EDU		0.102***		0.532***
		(0.012)		(0.077)
_cons	0.033***	0.001	0.029***	0.005
	(0.001)	(0.001)	(0.000)	(0.004)
N	1663	1676	2576	2822

Standard errors in parentheses.
*$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

Table 11
Mechanism test - crowding-out effect on S&T expenditure.

	(1)	(2)	(3)	(4)
	Resource-based Regions		Non- resource-based Regions	
	STI	GI	STI	GI
mining	−0.007**		−0.018	
	(0.003)		(0.023)	
STI		0.950***		6.957***
		(0.070)		(0.284)
_cons	0.002***	0.003***	0.002***	0.006***
	(0.000)	(0.001)	(0.000)	(0.002)
N	1663	1680	2576	2832

Standard errors in parentheses.
*$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

significant, indicating that resource dependence in non-resource-based cities does not have a negative impact on green technology innovation, that is, there is no crowding out effect.

In order to test the specific path of the "crowding out effect" of resource dependence on green technology innovation, the data regression test was performed on formula 4 and formula 5. The results in Table 10 reflect the Fixed Effects mechanism by which resource dependence has a crowding-out effect on human capital investment, thereby affecting the level of green technological innovation. In contrast, human capital investment has a significant promotional effect on green technology innovation. Therefore, the dependence on natural resources in resource-based areas will crowd out human capital investment, thus inhibiting green technological innovation and hindering green economic growth. This also verifies Hypothesis 2. From the sample of non-resource-based cities, resource dependence also appears to have a crowding-out effect on human capital investment. Therefore, both resource-based and non-resource-based cities should not be overly dependent on natural resources.

The regression results in Table 11 reflect the intrinsic mechanism of the crowding-out effect of resource dependence on science and technology R&D inputs, thus affecting the level of green technological innovation. The regression results show that regions that are dependent on natural resources see a crowding out of science and technology innovation inputs, thus inhibiting green technological innovation and hindering growth of the green economy, which also verifies Hypothesis 2. In the sample of non-resource-based cities, there is no similar crowding-out effect.

8. Conclusions and recommendations

8.1. Conclusions

This study verifies the inverted U-shaped relationship and regional heterogeneity between green technological innovation and growth efficiency of the green economy. It further tests the mechanism of impact between resource dependence, green technological innovation, and green economic growth in resource-based regions. Theoretically, this study clarifies the non-linear relationship between green technological innovation and growth efficiency of the green economy. It reveals the influence of the path of green technological innovation on green economic growth according to Muldaur's theory of cyclic causal accumulation and the theory of market failure. Practically speaking, this study establishes the importance of green economic growth, promoting the vigorous development of green technological innovation and deepening our understanding of how green economic growth is driven by technological innovation.

8.1.1 We thus draw the following conclusions. Green technological innovation and growth efficiency of the green economy are characterized by an "inverted U" relationship. When the level of green technology innovation is too low or too high, resource and non-resource cities experience less efficient green economic growth, indicating that there is a threshold level of green technological innovation or an effective interval of green technological innovation. Green technology innovation will increase the growth margin of green economy. When the development level of green technology innovation exceeds this range, the contribution rate of green technology innovation to the growth of green economy will decrease. For resource-based cities, when the technological innovation of the industries in the resource-processing sector enters the development bottleneck, the impact of green technological innovation on green economic growth will be inhibited.

8.1.2 The Heterogeneity analysis explores different types of resource cities. When the green technology innovation level of growth resource-based cities reaches a certain level, the marginal decreasing trend of growth efficiency of green economic becomes more obvious; the inverted U-shaped relationship is more significant, and their industries are mainly focused on resource extraction. With an increase in investment in green technological innovation, the expected output is reduced in the extended period after the transition, and green economic development may encounter a bottleneck. Mature resource cities are in a stable stage of resource development with resource security and appropriate infrastructure. At this time, an increase in investment in green innovation can promote green economic growth. Cities in decline or where there is resource regeneration due to the industrial structure and level of economic development are at a disadvantage, with little resource security and underdeveloped infrastructure for technological innovation infrastructure. As a result, the effect of green technological innovation on green economic growth is not significant.

8.1.3 Resource-based cities have squeezed out human capital investment and investment in science and technology innovation by investing excessive resources in resource-based sectors, thereby inhibiting green technology innovation and limiting green economic growth.

8.2. Recommendations for countermeasures

Based on the above findings, targeted countermeasures are proposed as follows.

8.2.1 The development of new industries and overcoming the "resource curse" should be encouraged, and homogenization of industrial structures should be halted. At the government level, policy guidance should be strengthened, and investment in green technology R&D should be increased. The government should strengthen its support for green technology innovation, especially in resource cities, and guide enterprises and research institutions to increase R&D of green technology and its application. Specifically, a special green technology R&D fund should be established to provide policy support, for example, through R&D loans and resource tax reform.

At the same time, enterprises should be encouraged to cooperate with universities and research institutes to promote the integrated development of industry, academic institutions, and research. The introduction of individual talent should be increased, and green awareness should be vigorously cultivated among in-house personnel. Enterprises should recruit and cultivate employees with green technology and environmental protection awareness to provide human resource security in the interests of promoting green economic growth.

Additional support for green innovation should be provided in other sectors. Many enterprises in resource cities invest considerable resources in non-resource-production sectors to maintain the stability of development, reducing green innovation capacity and inhibiting or reversing the growth of the green economy. Resources should be allocated equally among other sectors to promote green economic growth and avoid falling prey to the "resource curse." This will efficiently promote the healthy development of the government–industry–enterprise green-innovation ecosystem.

8.2.2 Differentiated environmental regulations should be implemented to eliminate the crowding out of green technological innovation in resource-based cities. Local governments should increase their expenditure on green technological innovation and simultaneously implement command-type environmental regulations for resource-based cities to effectively promote green technological innovation, reduce investment in resource-based industries, and steadily improve their green economic growth. It should furthermore encourage public participation in supervision, improve carbon and environmental taxation, develop emission permits and an emissions trading system, establish a system to supervise effective environmental protection and transform declining resource cities to regenerate and grow to complete their metamorphosis and advancement as mature resource cities.

8.2.3 Declining resource-based and non-resource-based cities with less developed levels of green technology innovation should actively introduce advanced green technology and management experience in those cities that are mature and resource-based.

Furthermore, provincial and regional institutional and policy barriers should be broken down. Independent green technology innovation should be promoted, a green technology standard should be developed, and the green industry should be upgraded. The government should speed up the formulation and improvement of green technology standards and the upgrading of the green industry in areas in which green innovation is less developed. Enterprises should be encouraged to adopt advanced green production technologies to reduce environmental pollution through the development of strict environmental protection standards and certification systems. At the same time, the government should transform traditional industries into green industries, thereby enhancing environmental protection.

8.2.4 The study on the relationship between green technology innovation and green economic growth not only provides guidance for the sustainable development of Chinese cities, but also brings valuable inspiration and lessons for other countries. First, resource-based cities should be deeply aware of the limited resources and the urgency of environmental protection. The traditional development model of over-reliance on resources is no longer sustainable, and green technology innovation has become the key to transformation and development. Resource-based cities in other countries should also draw lessons from this and plan for green transformation as soon as possible to avoid the dilemma of resource depletion and environmental deterioration. Secondly, non-resource-based cities should make full use of their industrial diversity and technological innovation advantages to promote the development of green industries. This experience is also applicable to other countries, especially in the context of globalization, non-resource-based cities should actively participate in international green economy cooperation and jointly promote the development of global green industries. International cooperation and exchanges play an important role in promoting green technological innovation and green economic growth. Countries should break down barriers, share green technology, experience and resources, and jointly tackle global environmental issues. Through international cooperation, the development and promotion of green technologies can be accelerated, and the overall level of the global green economy can be improved.

The realization of green economic growth needs to be driven by green technology innovation, and policy is an essential means to promote this innovation. The government should improve the research, development and application of green technologies and promote green economic growth by strengthening policy guidance, establishing a green technology standard, strengthening international cooperation, and cultivating those with skills in green technology. At the same time, enterprises should respond positively to policy guidance, adopt advanced green production technologies and management modes, and improve their environmental awareness and competitiveness. Only through the joint efforts of government and enterprises can the sustainable development of a green economy be realized.

Foundations

1. Natural Science Foundation of Inner Mongolia Project: Study on the Evolution and Regulation Mechanism of Resource and Environmental Carrying Capacity in Inner Mongolia (2021LHBS07001).

2. Basic Scientific Research Fund of Inner Mongolia Normal University: Study on the Evolution Characteristics and Efficiency of R&D Investment in Inner Mongolia (2022JBQN054).

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Data availability

The authors do not have permission to share data.

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